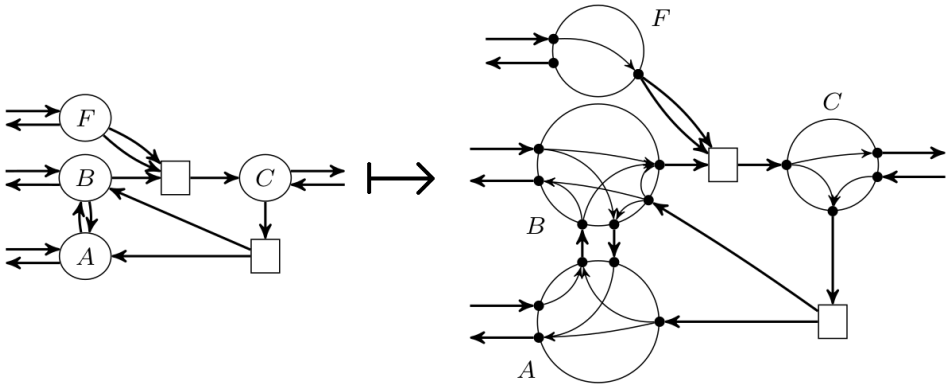
Layered Monoidal Theories

Leo Lobski (jww Fabio Zanasi)

University College London

9th International Conference on Applied Category Theory
Tallinn, 8 July 2026

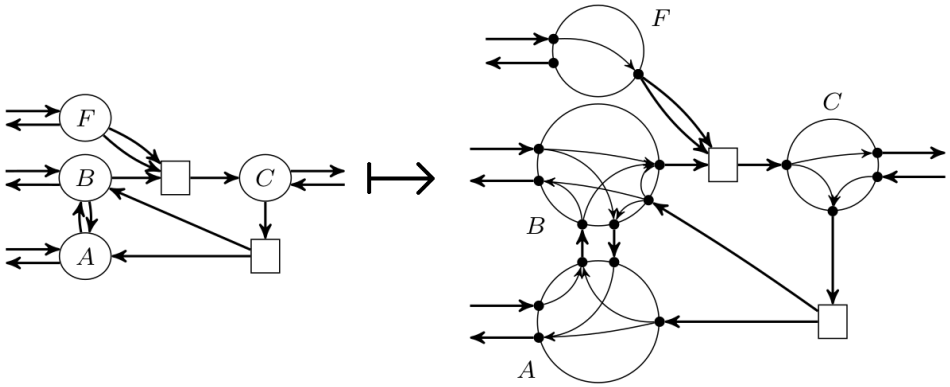
Motivation: layers of abstraction



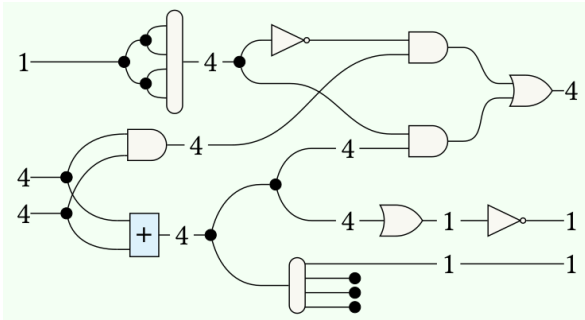
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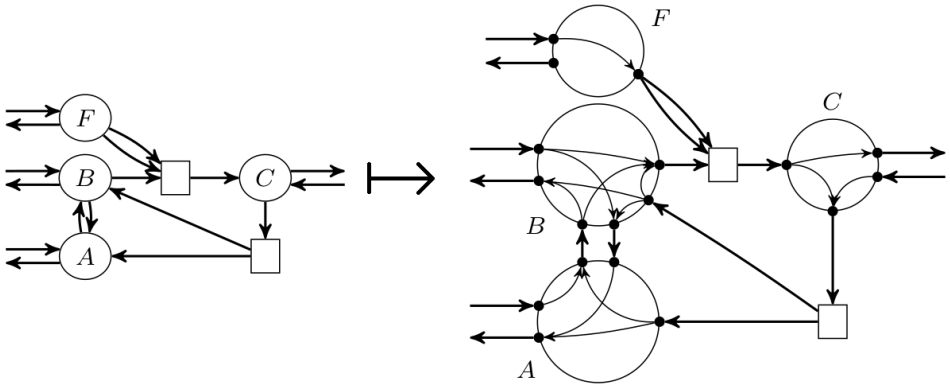


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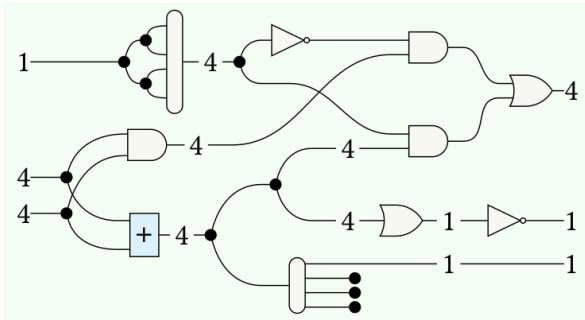
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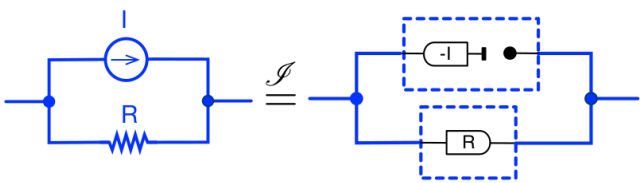
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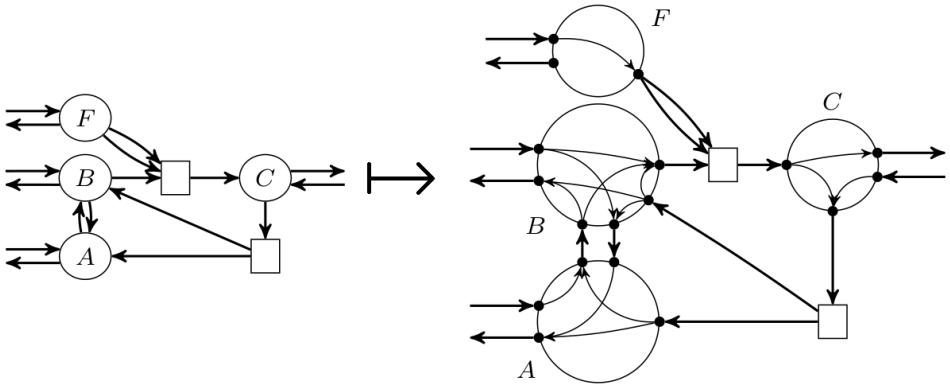
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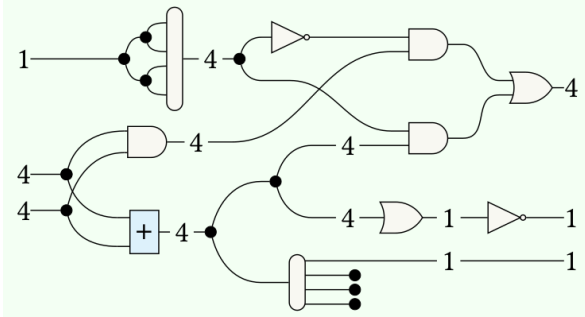
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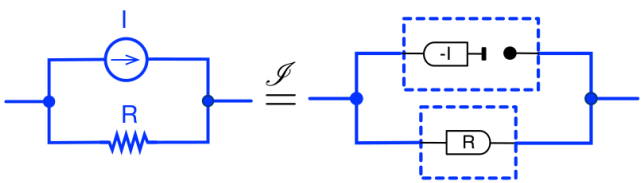
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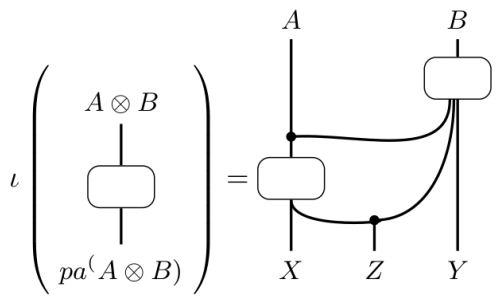
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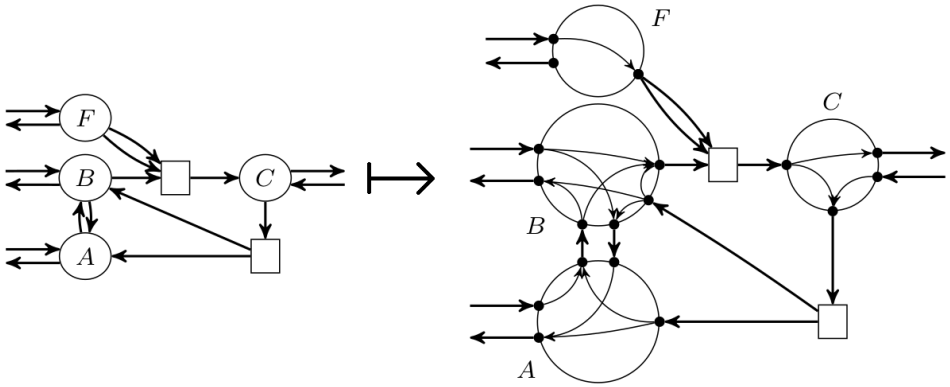
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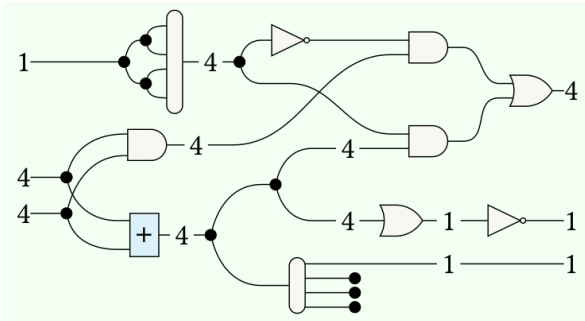
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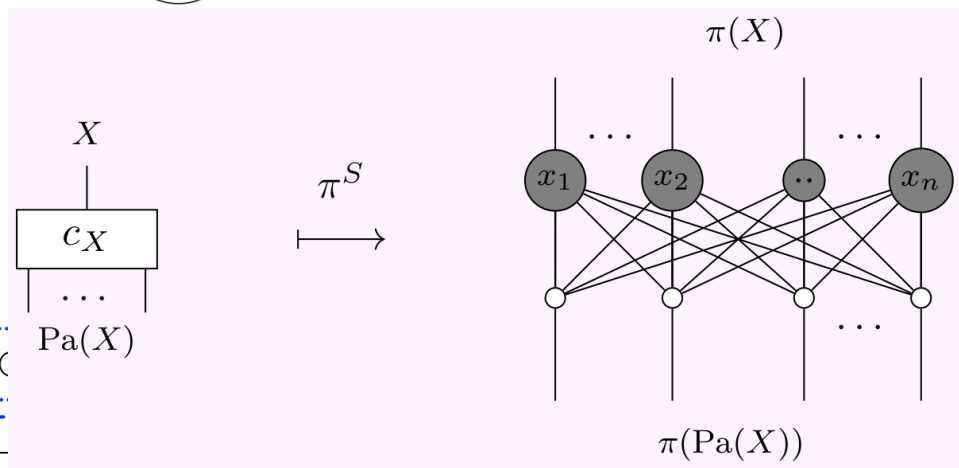
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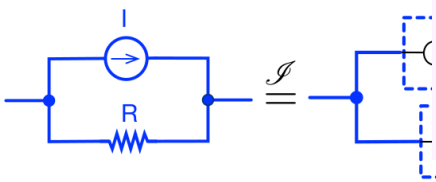
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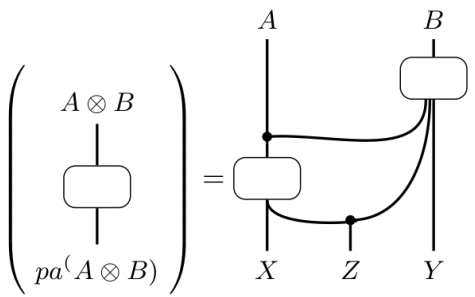
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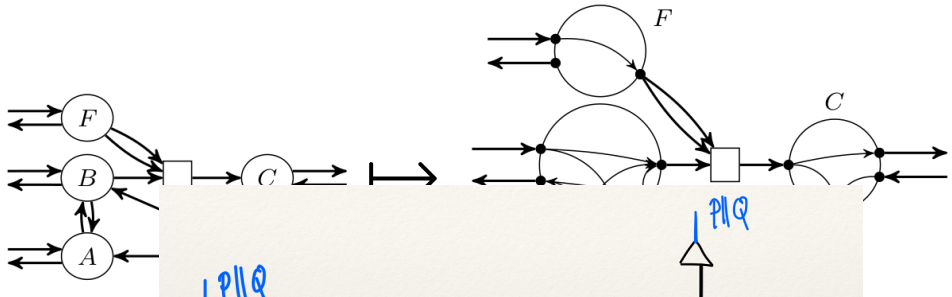
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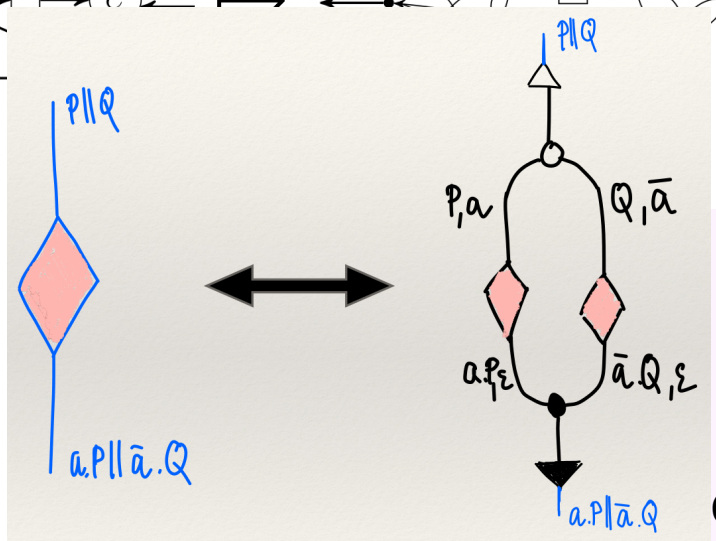
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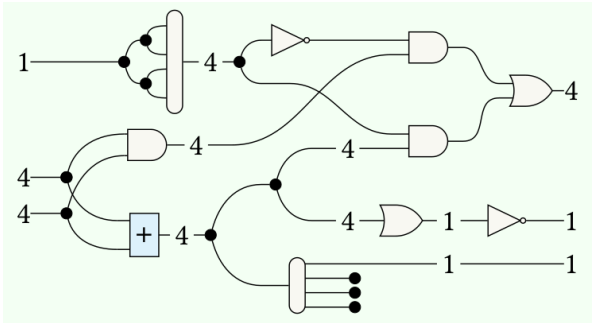
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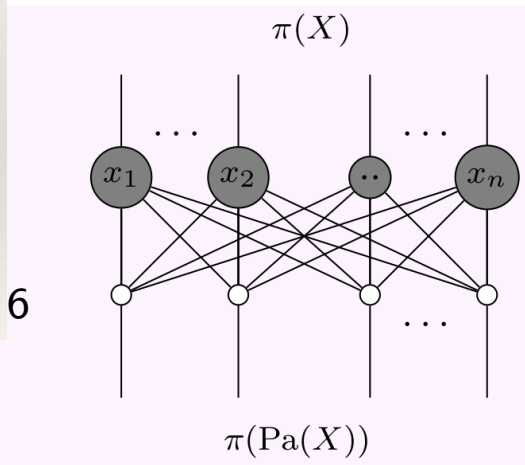
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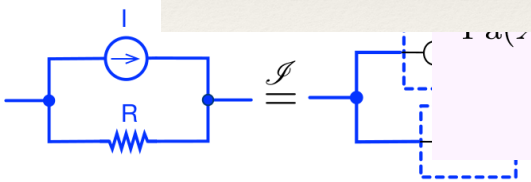
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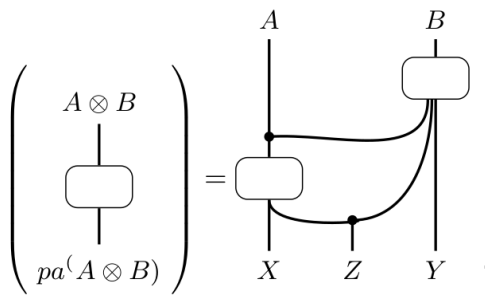


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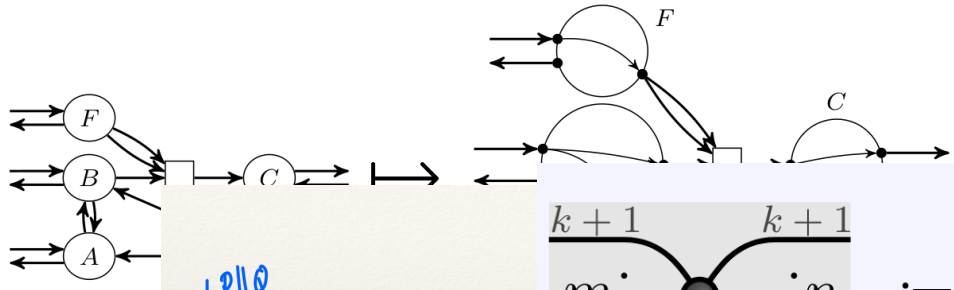
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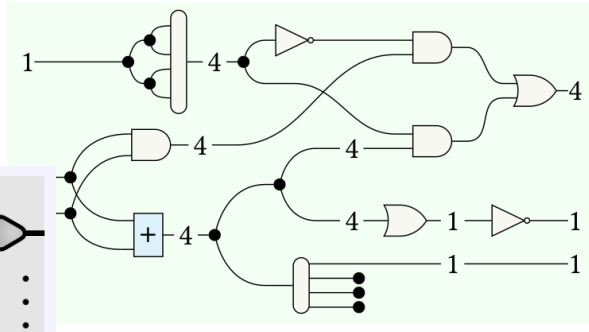
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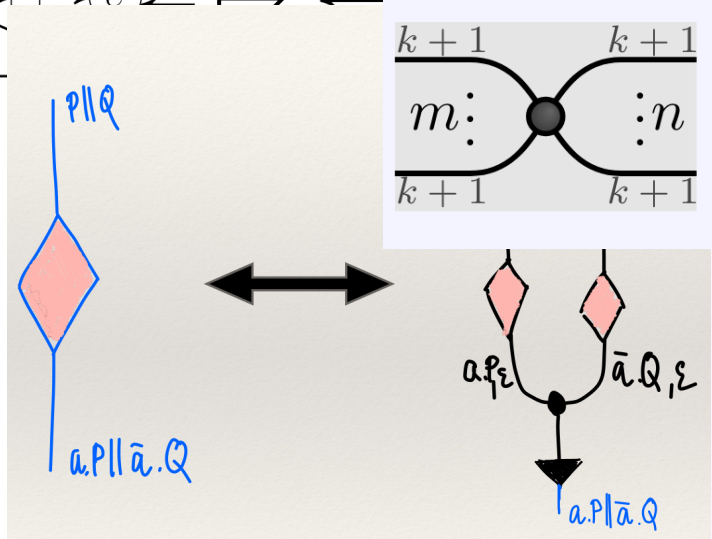
[6] Krivine. *Diagrammatic reasoning for physical systems and composite explanations*. 2019

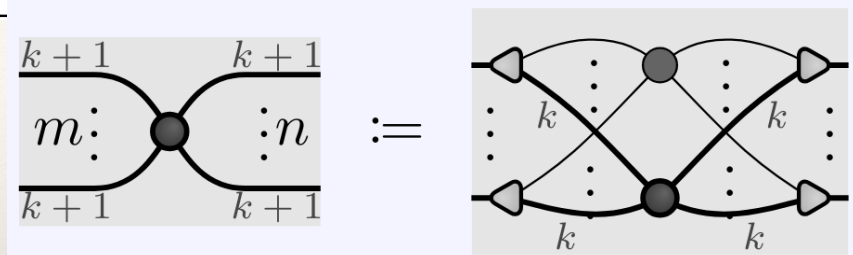
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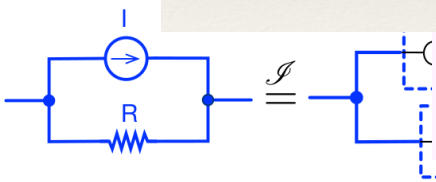
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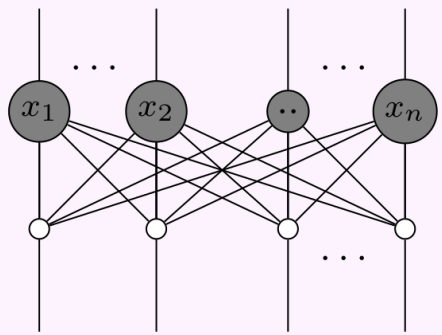




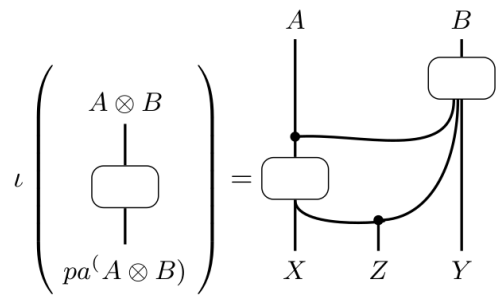
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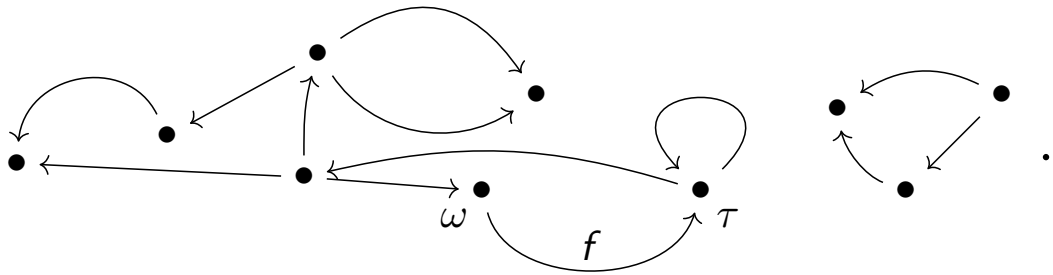
[7] Booth, Carette, Comfort. *Graphical Symplectic Algebra*. 2024

What is the purpose?

Wish to have a *syntactic* presentation for (strict) monoidal functors – akin to monoidal theories:

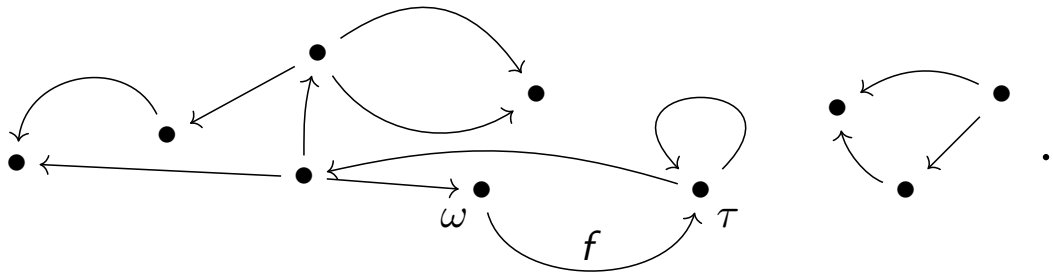
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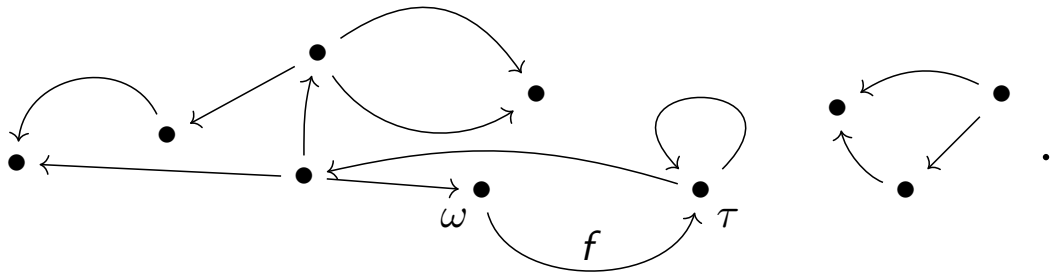
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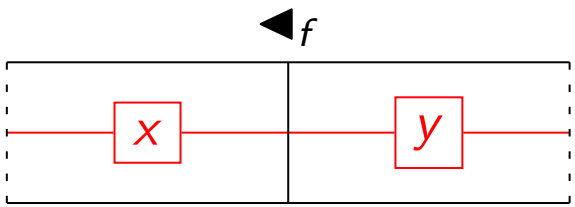
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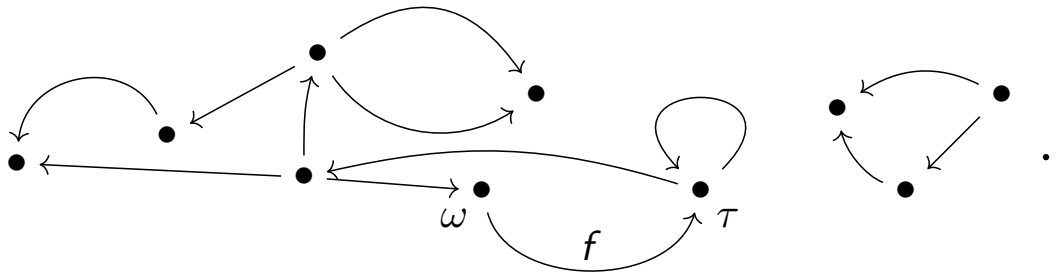


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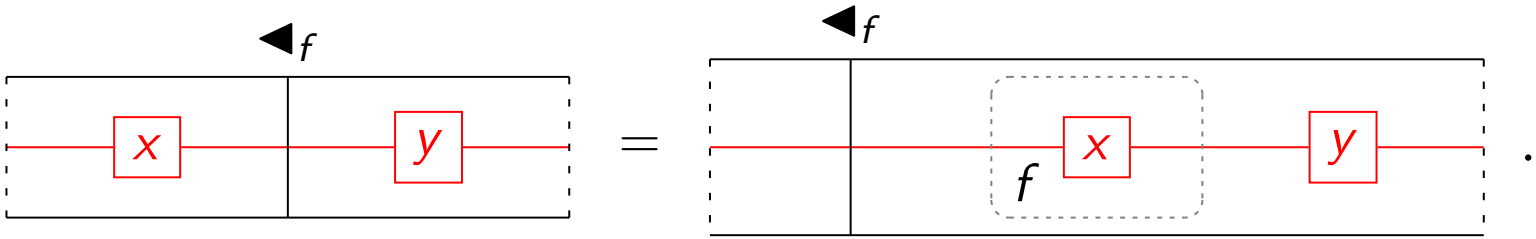


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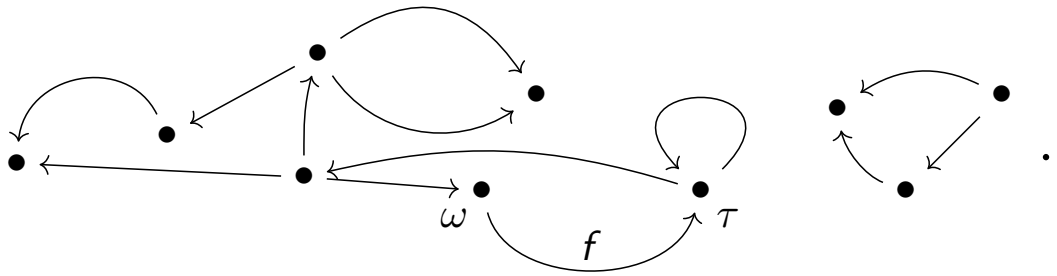


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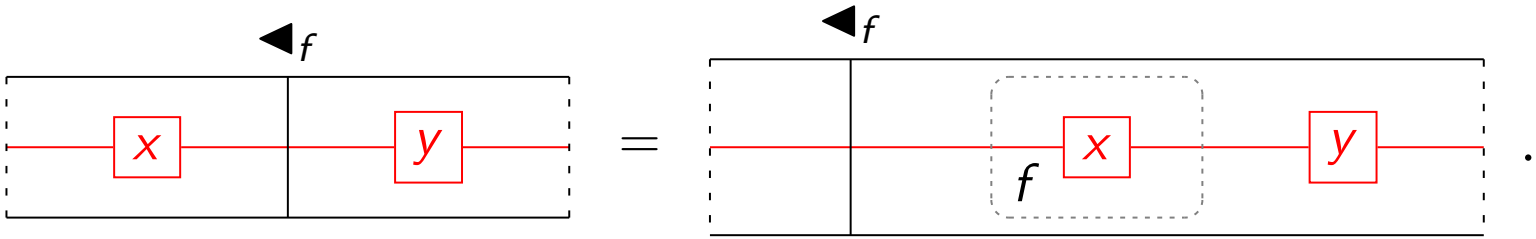


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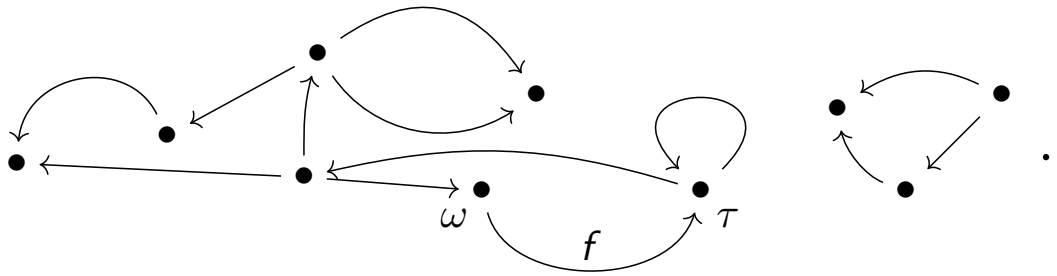
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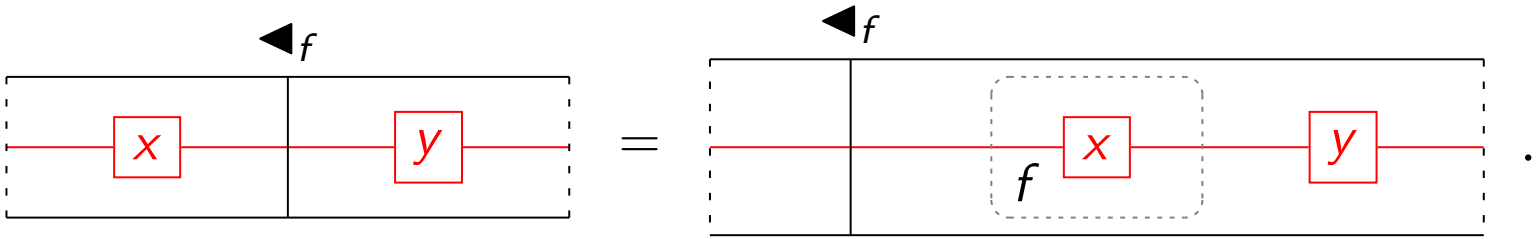
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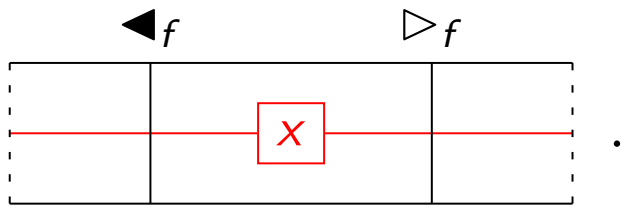
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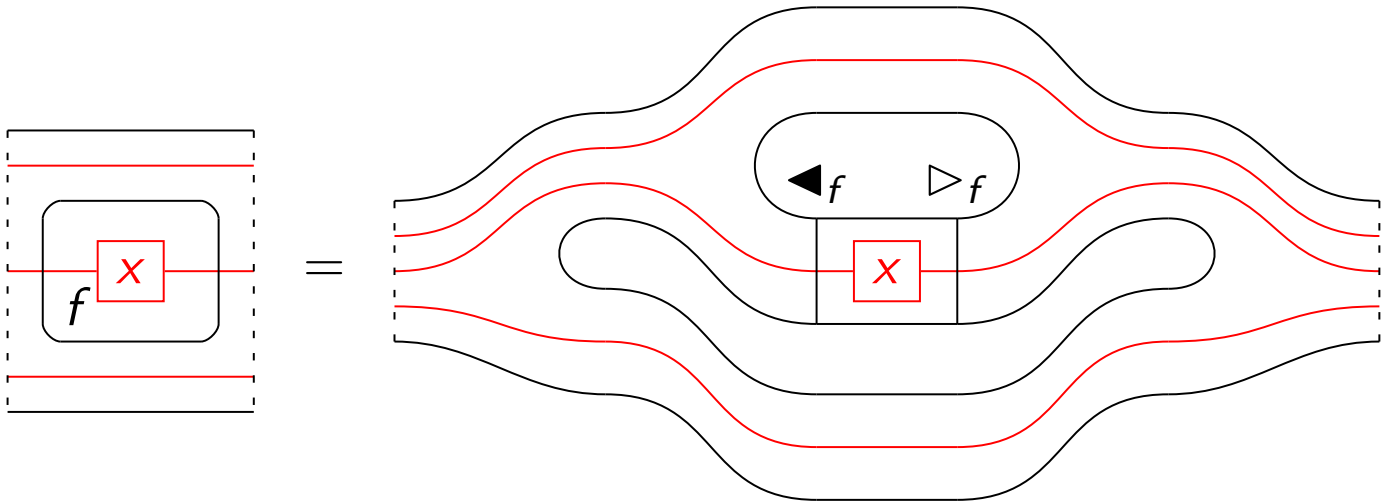


Functor coboxes

Bidirectional functor boundaries allow defining the *functor cobox*:

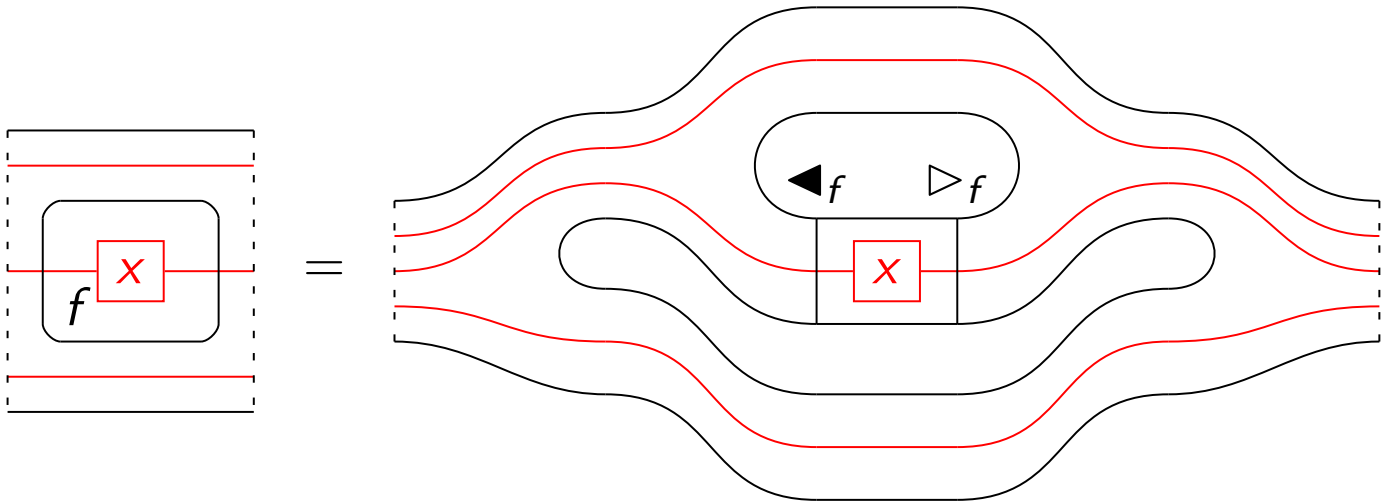
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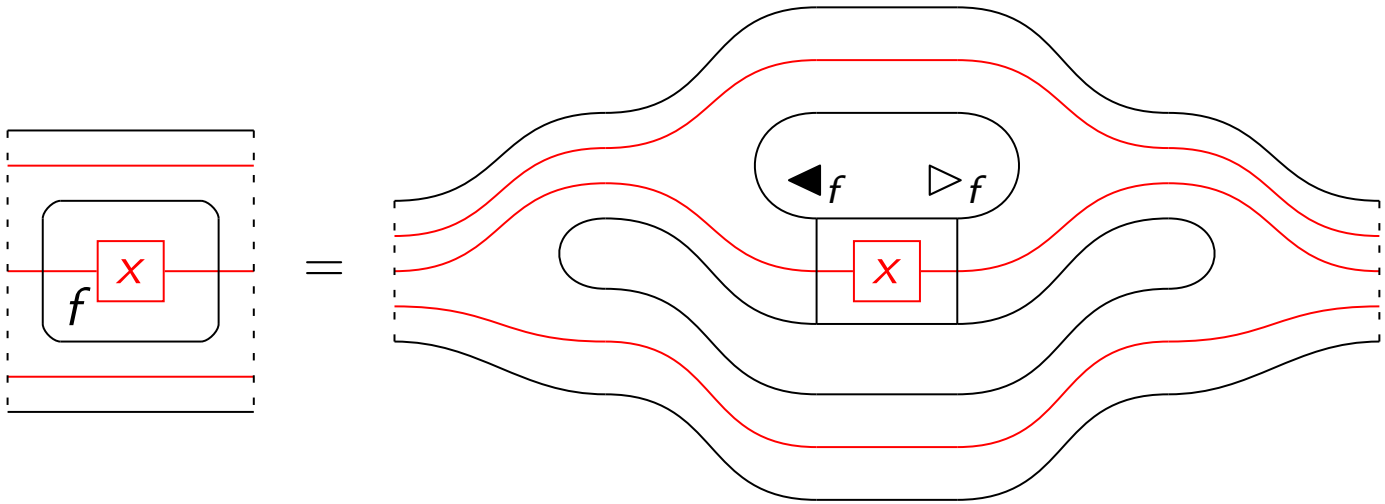
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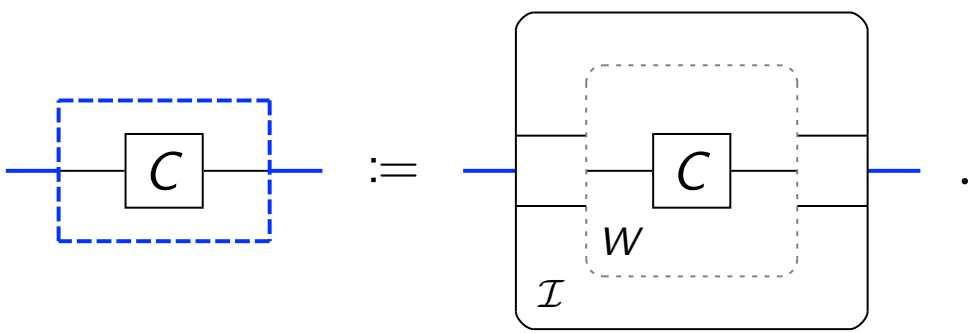
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- C_ω is a set of *colours*,
- $\Sigma_\omega : C_\omega^* \times C_\omega^* \to \mathbf{Set}$ is a function giving the *monoidal generators*.

Digital circuits: Signature

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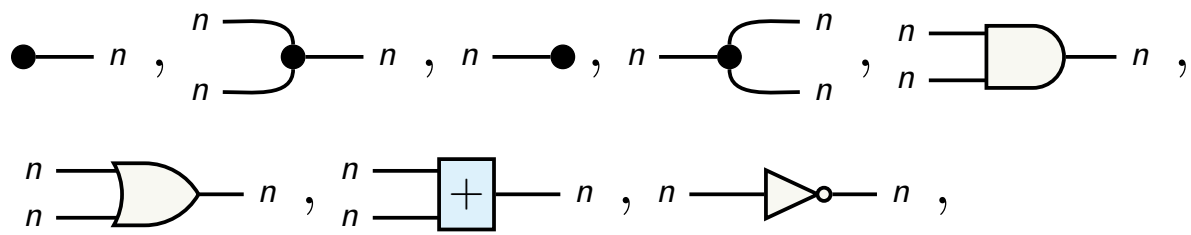
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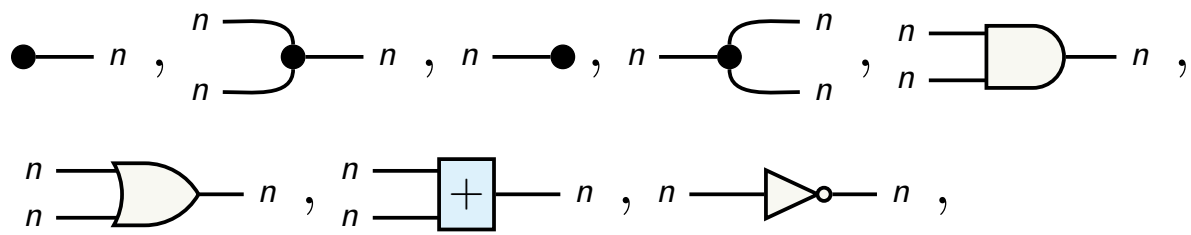
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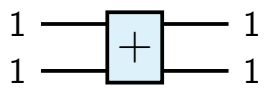


Digital circuits: Signature

Define the layered signature $(\mathbb{N}_+, \mathcal{F}, \Sigma_n)$ as follows:

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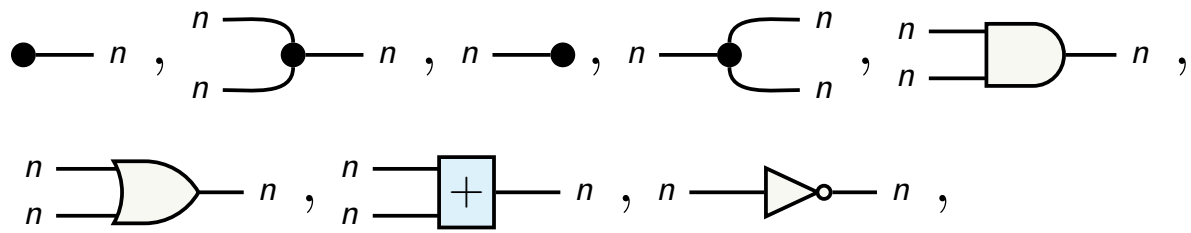
- for $n = 1$, the monoidal signature Σ_1 is as above, except that we replace the adder with 

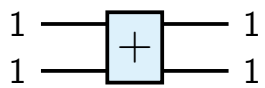
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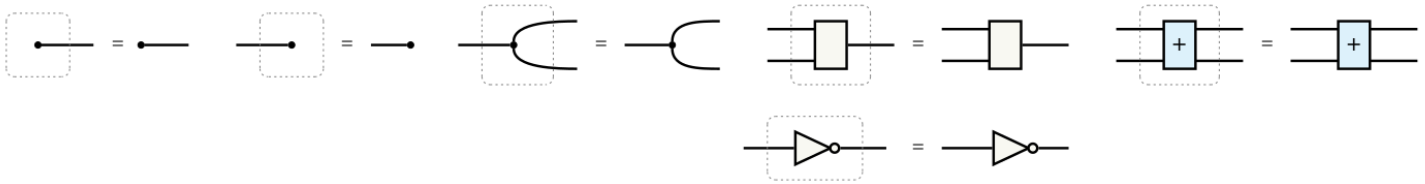
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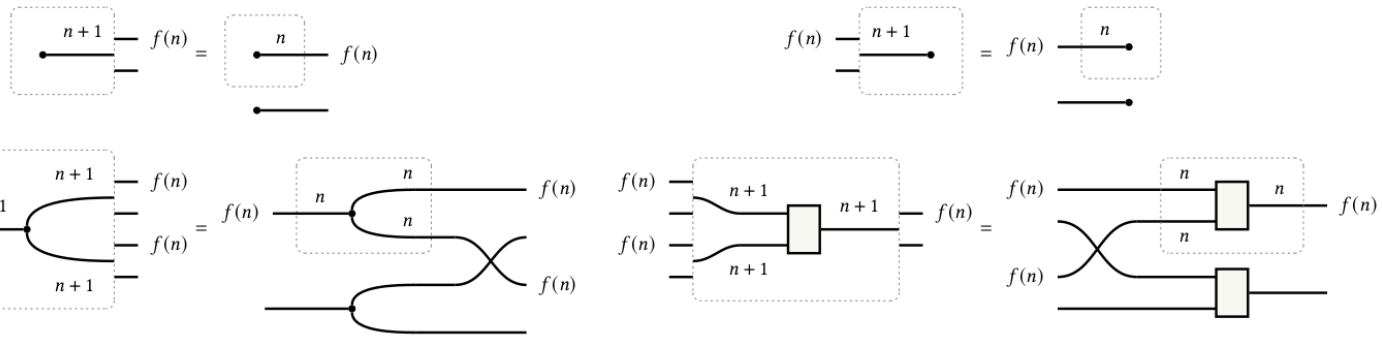
This signature is called *simple arithmetic circuits*.

Digital circuits: Theory

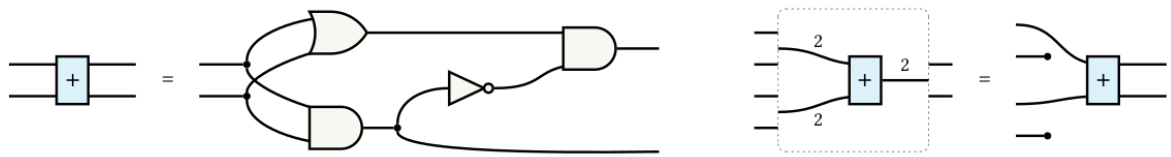
$f \in \mathcal{F}(1,1): \quad f(1) = 1$



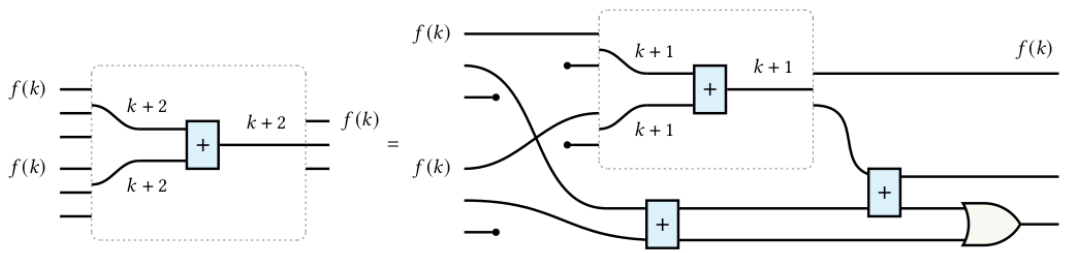
$f \in \mathcal{F}(n+1,1): \quad f(n+1) = f(n)1$

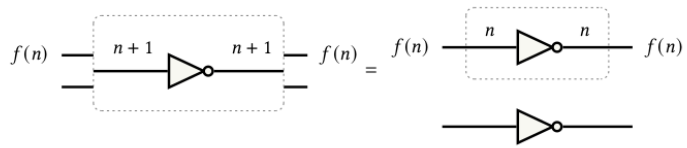


$n = 1:$

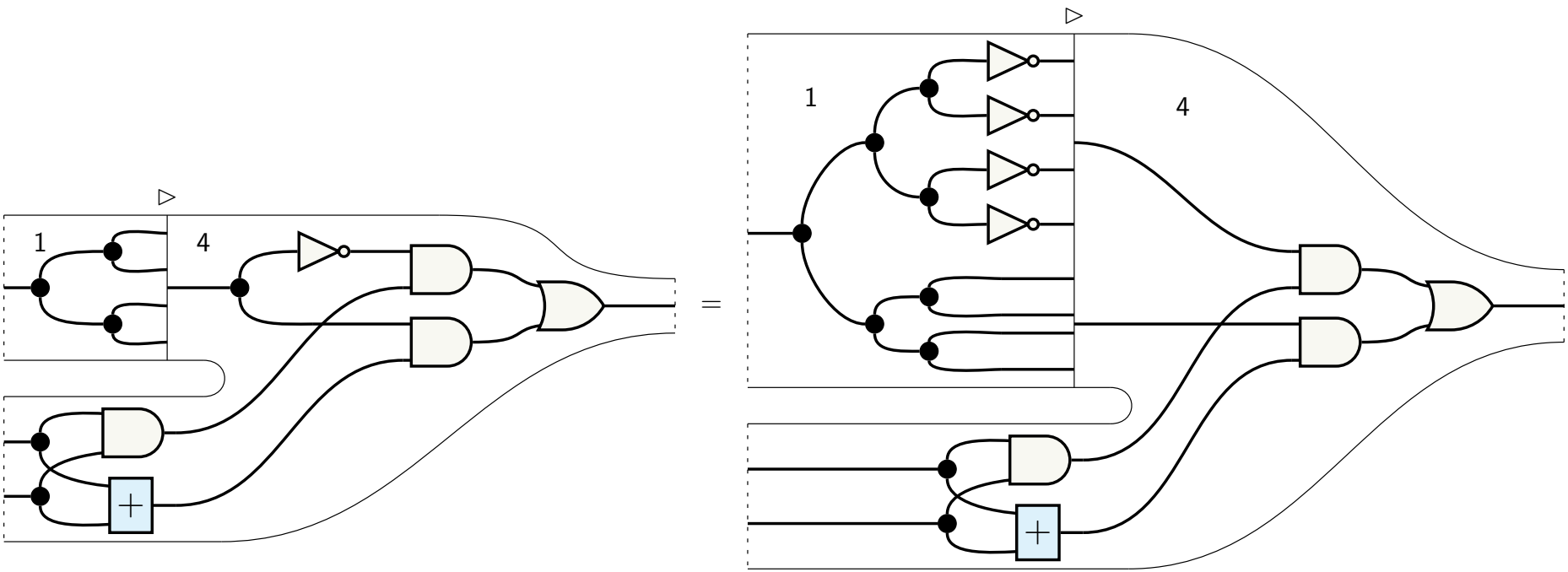


$n = k + 1$ for $k \geq 1:$





Digital circuits: A simple arithmetic logic unit



Semantics

theory	semantics (roughly)
monoidal theory	monoidal category
"forward" boundaries	monoid in the category of opfibrations
"backward" boundaries	comonoid in the category of fibrations
bidirectional boundaries	collage of certain diagram of profunctros

Related work

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- Holey string diagrams:
  - Matt Earnshaw and Mario Román. 2024. *Context-Free Languages of String Diagrams*.
  - James Hefford and Cole Comfort. 2023. *Coend Optics for Quantum Combs*.

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- L. and Fabio Zanasi. 2026. *Layered monoidal theories I: Diagrammatic algebra and applications*
- L. and Fabio Zanasi. 2026. *Layered monoidal theories II: Fibrational semantics*

Thank you for your attention!